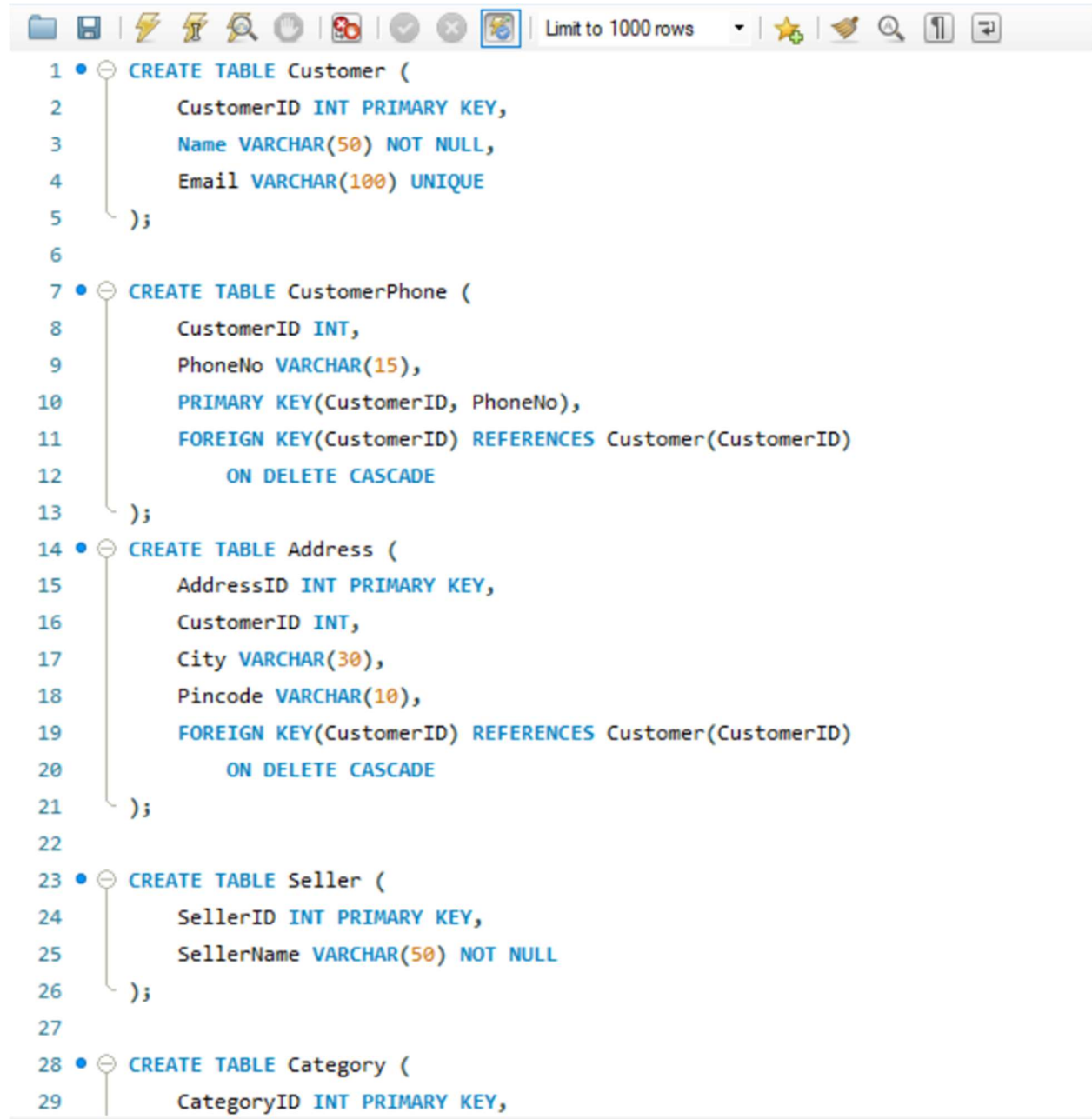


EXPERIMENT: 2

Aim:

To convert an ER diagram into a relational schema and implement it in MySQL using primary keys, foreign keys, NOT NULL, UNIQUE, ON DELETE CASCADE and ON DELETE SET NULL constraints. To insert sample data and demonstrate referential integrity.

Source Code:



```
1 • CREATE TABLE Customer (  
2     CustomerID INT PRIMARY KEY,  
3     Name VARCHAR(50) NOT NULL,  
4     Email VARCHAR(100) UNIQUE  
5 );  
6  
7 • CREATE TABLE CustomerPhone (  
8     CustomerID INT,  
9     PhoneNo VARCHAR(15),  
10    PRIMARY KEY(CustomerID, PhoneNo),  
11    FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)  
12        ON DELETE CASCADE  
13 );  
14 • CREATE TABLE Address (  
15     AddressID INT PRIMARY KEY,  
16     CustomerID INT,  
17     City VARCHAR(30),  
18     Pincode VARCHAR(10),  
19     FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)  
20        ON DELETE CASCADE  
21 );  
22  
23 • CREATE TABLE Seller (  
24     SellerID INT PRIMARY KEY,  
25     SellerName VARCHAR(50) NOT NULL  
26 );  
27  
28 • CREATE TABLE Category (  
29     CategoryID INT PRIMARY KEY,
```

```
30      CategoryName VARCHAR(50) UNIQUE
31    );
32 • ○ CREATE TABLE Product (
33     ProductID INT PRIMARY KEY,
34     ProductName VARCHAR(50) NOT NULL,
35     Price DECIMAL(10,2),
36     SellerID INT,
37     CategoryID INT,
38     FOREIGN KEY(SellerID) REFERENCES Seller(SellerID)
39       ON DELETE SET NULL,
40     FOREIGN KEY(CategoryID) REFERENCES Category(CategoryID)
41       ON DELETE SET NULL
42   );
43 • ○ CREATE TABLE ProductImage (
44     ProductID INT,
45     ImageURL VARCHAR(100),
46     PRIMARY KEY(ProductID, ImageURL),
47     FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
48       ON DELETE CASCADE
49   );
50
51 • ○ CREATE TABLE Physical_Product (
52     ProductID INT PRIMARY KEY,
53     Stock INT,
54     FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
55       ON DELETE CASCADE
56   );
57
58 • ○ CREATE TABLE Digital_Product (
```

```
57
58 • ○ CREATE TABLE Digital_Product (
59     ProductID INT PRIMARY KEY,
60     FileType VARCHAR(20),
61     FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
62       ON DELETE CASCADE
63   );
64 • ○ CREATE TABLE Orders (
65     OrderID INT PRIMARY KEY,
66     CustomerID INT,
67     AddressID INT,
68     OrderDate DATE,
69     FOREIGN KEY(CustomerID) REFERENCES Customer(CustomerID)
70       ON DELETE CASCADE,
71     FOREIGN KEY(AddressID) REFERENCES Address(AddressID)
72       ON DELETE SET NULL
73   );
74
75 • ○ CREATE TABLE OrderItem (
76     OrderID INT,
77     ProductID INT,
78     Quantity INT,
79     PRIMARY KEY(OrderID, ProductID),
80     FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
81       ON DELETE CASCADE,
82     FOREIGN KEY(ProductID) REFERENCES Product(ProductID)
83       ON DELETE CASCADE
84   );
85
```

```

85
86 • CREATE TABLE Payment (
87     PaymentID INT PRIMARY KEY,
88     OrderID INT UNIQUE,
89     Amount DECIMAL(10,2),
90     FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
91     ON DELETE CASCADE
92 );
93 • CREATE TABLE Delivery (
94     DeliveryID INT PRIMARY KEY,
95     OrderID INT UNIQUE,
96     Status VARCHAR(20),
97     FOREIGN KEY(OrderID) REFERENCES Orders(OrderID)
98     ON DELETE CASCADE
99 );
100
101
102
103
104 -- Sample Data
105
106 • INSERT INTO Customer VALUES
107     (1, 'Shreya', 'shreya@gmail.com');
108
109 • INSERT INTO CustomerPhone VALUES
110     (1, '9876543210');
111
112 • INSERT INTO Address VALUES
113     (101, 1, 'Lucknow', '226001');

```

```

112 • INSERT INTO Address VALUES
113     (101, 1, 'Lucknow', '226001');
114
115 • INSERT INTO Seller VALUES
116     (201, 'Tech World');
117
118 • INSERT INTO Category VALUES
119     (301, 'Electronics');
120 • INSERT INTO Product VALUES
121     (401, 'Laptop', 65000, 201, 301);
122
123 • INSERT INTO Physical_Product VALUES
124     (401, 10);
125
126 • INSERT INTO Orders VALUES
127     (501, 1, 101, '2026-08-19');
128
129 • INSERT INTO OrderItem VALUES
130     (501, 401, 1);
131
132 • INSERT INTO Payment VALUES
133     (601, 501, 65000);
134
135 • INSERT INTO Delivery VALUES
136     (701, 501, 'Shipped');
137 • INSERT INTO Orders VALUES
138     (502, 99, 101, '2026-08-19');
139 -- Display Output
140
141 • SELECT * FROM Customer;
142 • SELECT * FROM Product;
143 • SELECT * FROM Orders;

```

OUTPUT:

Result Grid



Filter Rows:

	CustomerID	Name	Email
▶	1	Shreya	shreya@gmail.com
✱	NULL	NULL	NULL

Result Grid		Filter Rows:		Edit:	
	ProductID	ProductName	Price	SellerID	CategoryID
▶	401	Laptop	65000.00	201	301
*	NULL	NULL	NULL	NULL	NULL

Result Grid



Filter Rows:

	OrderID	CustomerID	AddressID	OrderDate
▶	501	1	101	2026-08-19
•	NULL	NULL	NULL	NULL

Result Grid




Filter Rows:

	OrderID	ProductID	Quantity
▶	501	401	1
•	NULL	NULL	NULL

Result:

The given ER diagram was successfully converted into a relational schema and implemented in MySQL using Primary Key, Foreign Key, NOT NULL, UNIQUE, ON DELETE CASCADE, and ON DELETE SET NULL constraints. Sample data was inserted and referential integrity was successfully demonstrated using a foreign key violation.